



Technical Service Bulletin

Technical Service Bulletin: TSB210107

Released Date: 12-Jan-2026

Engine Wiring Harness Approved Repairs

Engine Wiring Harness Approved Repairs

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

All Products where a braided wiring harness is subjected to fluid ingress of either environmental, operational, or maintenance procedures, such as on site cleaning operations. Or, where a harness sheathing has been damaged and the internal wiring is not compromised.

- QSK38 CM2150 K106
- QSK38 CM2150 K142M
- QSK38 CM2350 K125
- QSK38 CM2350 K138M
- QSK38 CM2350 K145G
- QSK50 CM2150 K107
- QSK50 CM2150 K140M
- QSK50 CM2350 K108
- QSK50 CM2350 K130
- QSK50 CM2350 K146G
- QSK60 CM2150 MCRS
- QSK60 CM2150 K142M
- QSK60 CM2350 K112
- QSK60 CM2350 K116
- QSK60 CM2350 K117
- QSK60 CM2350 K121
- QSK60 CM2350 K135
- QSK60 CM2350 K136
- QSK60 CM2350 K149M
- QSK78 CM2350 K126
- QSK78 CM2350 K137

Issue

Fluid Ingress and Corrosion on engine wiring harness connectors.

Damaged wiring harness breakouts where the sleeving or covering has been damaged and the wiring has not been compromised.

Symptom

- Multiple warning fault codes are present (depending on which circuits were affected)
- Visual inspections reveal splits, tears, perforations or fretted harness coverings.

Root Cause

- Fluid can ingress into the wiring loom due to breaks in the main loom coverings and failed covering seals and bypass the internal seal of the engine wiring harness connector.

Verification

- Verify presence of fluids or a blue or green residue/salts in the engine wiring harness connectors as shown below in Figure 1.

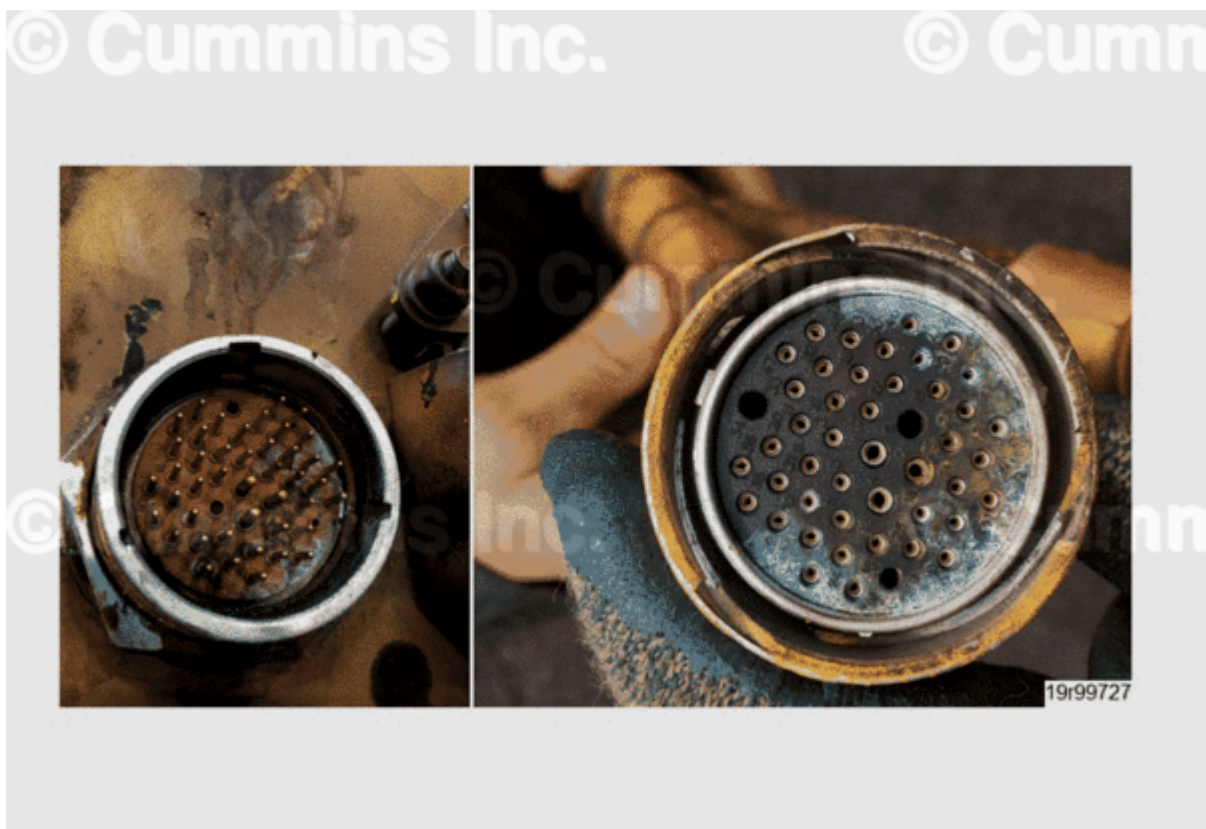


Figure 1, Engine Wiring Harness Connector Corrosion.

Resolution

- This document outlines actions to be taken for engine wiring harness corrosion related issues or for proactive peace of mind fixes.
- Avoid tearing the harness down. It is best to leave the existing harness protection in place. Disconnect and upend the affected harness connector, giving it a good shake to allow trapped fluid to become displaced. Use an airline to blow out any visual fluids.
- See Service Instruction section below for details on how to apply recommended countermeasures to existing engine wiring harnesses to help reduce fluid ingress and corrosion.

- Individual wiring harnesses that are released with improvements will be released under separate a TSB.

Service Instruction

- The preventative action instructions below focus on the rear crossover connections but can be applied to any connector covering.

Recommended Materials

- The following products in Table 1 below are recommended for use when performing preventative or repair actions to harness connectors and or coverings.

Note : If the items below are not available in your area/region, a similar product is acceptable. Make sure the products used are able to be subjected to temperatures of up to 120 °C

Table 1, Recommended Materials

Item (Harness)	Item	Specific Product
Conduit	Heat Shrink	TE Connectivity RP-4800 Series Heat Shrink Tubing and Sleeves
	Cold Shrink	GremTek GREMTUBE® GCSTS
		Silicone cold shrink tubing 3M™ Cold Shrink Tubing
	Tape/Wrap	Techspan Wrap & Seal Rubber tape
		X-Treme Tape ® Self-Fusing Silicone Rubber Tape
Braid	Tape/Wrap	Techspan Wrap & Seal Rubber tape
		X-Treme Tape ® Self-Fusing Silicone Rubber Tape
Heat Shrink	Tape/Wrap	Techspan Wrap & Seal Rubber tape
		X-Treme Tape ® Self-Fusing Silicone Rubber Tape
Connector (Fitted)	Cold Shrink	GremTek GREMTUBE® GCSTS
		Silicone cold shrink tubing 3M™ Cold Shrink Tubing
	Tape/Wrap	Techspan Wrap & Seal Rubber tape
		X-Treme Tape ® Self-Fusing Silicone Rubber Tape

Table 1, Recommended Materials		
Item (Harness)	Item	Specific Product
Connector (Unfitted)	Corrosion Prevention Compound	Corrosion X

Rear Crossover Connection Corrosion Prevention (Conduit)

- Check the condition of the affected wiring harness before starting this procedure. If corrosion is present in the connector, clean it out as best as possible. If the engine has flagged fault codes that could be associated with corrosion, replace the affected wiring harness and carry out this procedure on the replacement part. If no fault codes are present, proceed with the instructions below
- Disconnect the rear engine wiring harness crossover connector.
- Clean both the female and male side of the crossover connector thoroughly using an electrical contact cleaner or similar and dry with compressed air.
- Install both ends of the engine wiring harness section on the engine. Once installed, correctly mark the angled end to the loom for the proper orientation as shown in Figure 2 below.
- Once marked, if possible, remove the engine wiring harness and place on a bench.

Note : This step may only apply to some engine models.

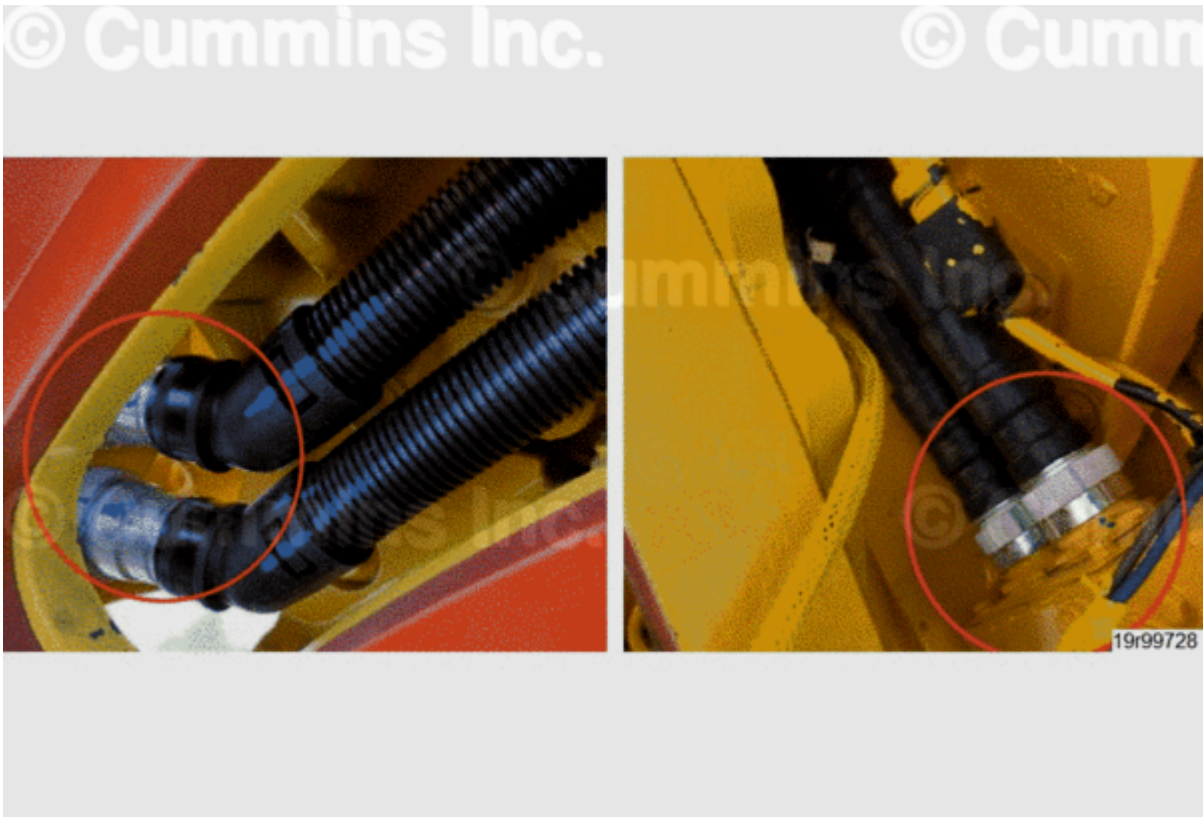


Figure 2, Showing Engine Wiring Harness Connector Marking.

- Measure and mark approximately 15 cm [6 in] from the engine wiring harness connector and apply a wrap of rubberized tape. Make a wrap of approximately 3 cm [1 in] wide and verify orientation is similar to Figure 3 below. The tape should self-adhere. The tape is added

as a precaution to help seal the heat shrink / cold shrink added later. The tape will also help fill the voids of the harness conduit.



Figure 3, Rubber Tape Application.

- Take the heat or cold shrink and slide over the engine wiring harness connector. Verify heat or cold shrink is kept as close as possible to the engine wiring harness connector. Cut an appropriate length of wrap, verify the wrap is long enough to cover past the applied rubberized tape.
- If using heat shrink, use a heat gun to start the shrinking process. Start the process at the engine wiring harness connector end, verifying that as much shrink tube as possible is kept on the connector end and slowly move down the rest of the wrap until all shrink tube is completed. See Figure 4 below.

⚠ CAUTION ⚠

Do not overheat or over shrink the wrap as this can cause splitting.



Figure 4, Finished Heat Shrink Wrap.

- If using cold shrink wrap, application may be easier with the connector installed. If this case, install the connector, hold the cold shrink flush against the connector body carefully to **not** interfere with the rotating part of the connector. See Figure 5 below.
- Pull on the plastic tab so the cold shrink is applied. Note the direction of the pull-tab when using cold shrink. Cold shrink should be pulled from the front and through the cold shrink itself towards the back.

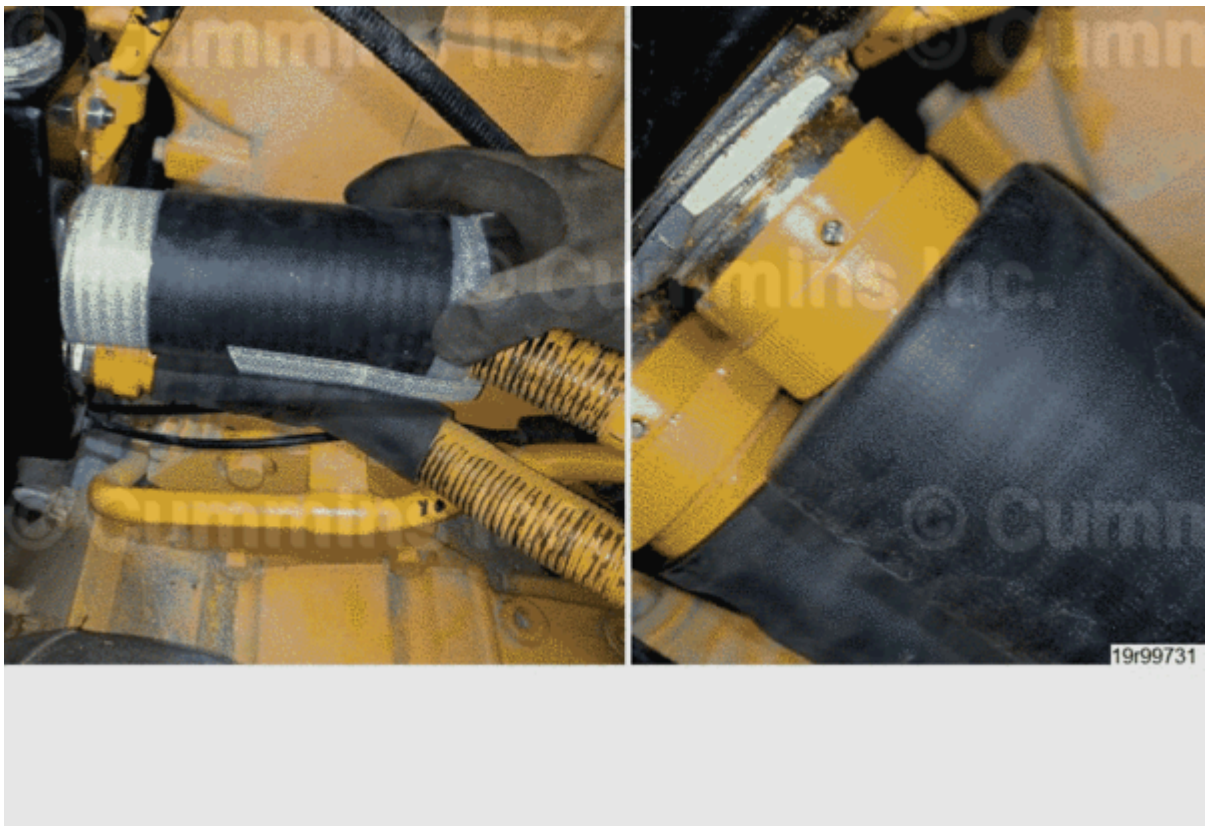


Figure 5, Cold Shrink Mating.



Figure 6, Cold Shrink Application.

- In the application of either heat or cold shrink, verify that the wrap is applied as close to the rotating part of the connector as possible. This will ensure the best seal with the wrap.



Figure 7, Completed Wrap Installation.

- Repeat the above steps for all necessary engine wiring harness connectors, adjusting the wrap length and diameters as necessary for the connectors being sealed.
- Before connecting the engine wiring harnesses connectors, a corrosion prevention compound (such as Corrosion X) should be applied to the connectors before mating. Apply a light layer of the compound on both the male and female connectors. If this compound is not available in your region, a non-silicone base di-electric grease can be used as a substitute. Apply a light layer of di-electric grease to the male side connector pins before assembling.

Heat Shrunk Coverings

Heat Shrunk Loom Covering Repairs and or Preventative Actions

Small cuts, splits or perforations to the coverings can be repaired by using the specified Tape/Wrap by covering the damage entirely including overlap of +30mm each side of the damaged area.

Braided Coverings

Small cuts, splits or perforations to the coverings can be repaired by using the specified Tape/Wrap by covering the damage entirely including overlap of +30mm each side of the damaged area.

Fluid Ingress

Mark the as installed P-Clip position with a paint pen line to be able to repeat position, rotation and orientation of the P-Clip. Disconnect and remove the affected harness from the engine and remove any brackets and ties. Upend the harness by the main connector and shake to back flush any trapped fluid in the main connector and use an airline to dry/remove any visible fluid. Do not remove the existing protective covering. Tape/wrap the dried harness from and over the connector backshell to around 20mm short of the first feature this will most likely be a P-Clip or a breakout...

P-Clip: Position the P-Clip to its installed marker and continue wrapping/taping the harness over the P-Clip. Ensure no gaps exist in the taping.

Breakout: If the harness presents a breakout, carry on taping/wrapping along the main loom and tape/wrap the breakout junction afterwards. Ensure no gaps exist in the taping.

With the harness fully taped/wrapped reinstall any removed brackets and ties and reinstall the harness to the engine. It is also possible to apply tape/wrap to the connector when fully installed if there is sufficient access.

Document History

Date	Details
2021-5-14	Module Created
2021-9-9	Added Product Affected section.
2023-11-14	Added QSK38 SMNs to Product Affected list.
2025-12-18	Non-Product Problem Solving (PPS)